



Multiple Choice:

1. A. (34/36)(600 Hz)

Solution:

Since the car is moving AWAY from the observer, use the formula for Doppler Effect (away):

$$f' = f \frac{v - v_o}{v + v_s} = (600) \frac{340 - 0}{340 + 20}$$

$$f' = (600) \frac{34}{36} \rightarrow \text{Ans}$$

Reference: GEAS 02-11 – The Doppler Effect

2. D. 20 dB

Solution:

Intensity is inversely proportional to the distance squared: $I = \frac{P}{A} = \frac{P}{4\pi r^2}$

This means that if the distance increases by a factor of 10, the intensity decreases by a factor of 10^2 or 100.

$$I = 10 \log \frac{1}{100} = -20 \text{ dB} \rightarrow \text{Ans}$$

Reference: GEAS 02-10 – Sound

3. B. 1500 m/s

Solution:

$$l = \frac{v}{f}$$

$$3 \text{ m} = \frac{v}{500 \text{ Hz}}$$

$$v = 1500 \frac{\text{m}}{\text{s}} \rightarrow \text{Ans}$$

Reference: GEAS 02-10 – Sound

4. A. 1 m

Solution:

Speed of sound in air at room temperature is roughly 343 m/s.

$$l = \frac{v}{f} = \frac{343 \frac{\text{m}}{\text{s}}}{343 \text{ Hz}} = 1 \text{ m} \rightarrow \text{Ans}$$

Reference: GEAS 02-10 – Sound

5. C. $t = 2 \text{ s}$

Solution:

$$l = \frac{v}{f} = vT$$

$$8 \text{ m} = \left(2 \frac{\text{m}}{\text{s}}\right) T$$

$$T = 4 \text{ s}$$

If the period of the wave is 4 s, the time from the crest to trough must be 2 s (or half of the period)

Reference: GEAS 02-09 – Waves

6. C. 5 m

Solution:

$$\lambda = \frac{v}{f} = \frac{\sqrt{\frac{T}{\mu}}}{f} = \frac{\sqrt{\frac{50 \text{ N}}{0.5 \frac{\text{kg}}{\text{m}}}}}{2 \text{ Hz}} = 5 \text{ m} \rightarrow \text{Ans}$$

Reference: GEAS 02-09 – Waves

7. C. 20 m/s

Solution:

$$v = \sqrt{\frac{T}{m}} = \sqrt{\frac{80 \text{ N}}{\frac{1 \text{ kg}}{5 \text{ m}}}} = 20 \frac{\text{m}}{\text{s}} \rightarrow \text{Ans}$$

Reference: GEAS 02-09 – Waves

8. D. 6 cm, inverted (correction, should be 2 cm)

Solution

$$\frac{1}{f} = \frac{1}{i} + \frac{1}{o}$$

$$\frac{1}{10 \text{ cm}} = \frac{1}{i} + \frac{1}{30 \text{ cm}}$$

$$i = +15 \text{ cm (real)}$$

$$m = -\frac{i}{o} = \frac{h_i}{h_o}$$

$$-\frac{15}{30} = \frac{h_i}{4 \text{ cm}}$$

$$h_i = -2 \text{ cm or } 2 \text{ cm inverted}$$

Reference: GEAS 02-13 – Mirrors

9. A. 208 nm

Solution:

$$n_{\text{diamond}} = \frac{c}{v_{\text{diamond}}}$$

$$2.4 = \frac{3 \times 10^8 \frac{\text{m}}{\text{s}}}{v_{\text{diamond}}}$$

$$v_{\text{diamond}} = 125 \times 10^6 \frac{\text{m}}{\text{s}}$$

Get the frequency of light in air:

$$l_{\text{air}} = \frac{v_{\text{air}}}{f}$$

$$500 \text{ nm} = \frac{3 \times 10^8 \frac{\text{m}}{\text{s}}}{f}$$

$$f = 6 \times 10^{14} \text{ Hz}$$

The frequency of light in air will be the same as the frequency of light in diamond (frequency will never change)

$$l_{\text{diamond}} = \frac{v_{\text{diamond}}}{f}$$

$$l_{\text{diamond}} = \frac{125 \times 10^6 \frac{\text{m}}{\text{s}}}{6 \times 10^{14} \text{ Hz}} = 208 \text{ nm} \rightarrow \text{Ans}$$

Reference: GEAS 02-12 – Nature of Light

10. D. Image is real, 120 cm away

Solution:

$$\frac{1}{f} = \frac{1}{i} + \frac{1}{o}$$

$$\frac{1}{40 \text{ cm}} = \frac{1}{i} + \frac{1}{60 \text{ cm}}$$

$$i = +120 \text{ cm (real)}$$

Reference: GEAS 02-13 – Mirrors

11. B. $3.0 \times 10^{-10} \text{ m}$

Solution:

$$l = \frac{c}{f} = \frac{3 \times 10^8 \frac{\text{m}}{\text{s}}}{1 \times 10^{18} \text{ Hz}}$$

$$l = 3 \times 10^{-10} \text{ m} \rightarrow \text{Ans}$$

Reference: GEAS 02-12 – Nature of Light



12. 37.5 cm (correction, not in the choices)

Solution:

$$m = -\frac{i}{o} = \frac{h_i}{h_o}$$

$$-2 = -\frac{i}{o}$$

$$i = 2o$$

$$\frac{1}{f} = \frac{1}{i} + \frac{1}{o}$$

$$\frac{1}{25 \text{ cm}} = \frac{1}{2o} + \frac{1}{o}$$

$$o = 37.5 \text{ cm} \rightarrow \text{Ans}$$

Reference: GEAS 02-13 – Mirrors

13. C. 40 cm

Solution:

$$\frac{1}{f} = \frac{1}{i} + \frac{1}{o}$$

$$\frac{1}{20 \text{ cm}} = \frac{1}{i} + \frac{1}{40 \text{ cm}}$$

$$i = 40 \text{ cm} \rightarrow \text{Ans}$$

Reference: GEAS 02-14 – Lenses

14. B. It cannot form real images.

Solution:

Concave lenses always produce virtual and upright images,

Reference: GEAS 02-14 – Lenses

15. B. 30° and 119°

Solution:

Reflection: If the angle of incidence (with respect to the normal) is 30 degrees, then the angle of reflection is also 30 degrees. Note that the 60-degree angle given in the problem is with respect to the surface, not the normal line.

Refraction: Use Snell's Law

$$n_1 \sin \theta_i = n_2 \sin \theta_r$$

$$1.0 \sin 30^\circ = 1.5 \sin \theta_r$$

$$\theta_r = 19.47^\circ \rightarrow \text{Ans}$$

Reference: GEAS 02-12 – Nature of Light

16. A. 27 cm from the second lens

Solution:

The image position for the first lens is d_{1i} where $1/25 \text{ cm} = 1/6 \text{ cm} + 1/d_{1i}$. $d_{1i} = -7.9 \text{ cm}$ (virtual image). Thus the object distance for the second lens is $d_{2o} = 9 \text{ cm} + 7.9 \text{ cm} = 15.9 \text{ cm}$.

$$\frac{1}{10 \text{ cm}} = \frac{1}{15.9 \text{ cm}} + \frac{1}{d_i}$$

$$d_i = 27 \text{ cm} \rightarrow \text{Ans}$$

The final image (real) is thus 27 cm from the second lens and 41 cm from the bug.

Reference: GEAS 02-14 – Lenses

17. C. 11.7 s

Solution:

Note: Solve first for the speed of sound in air at 20°C. The speed of sound in air at 0°C is approximately 330 m/s

$$v = 330 \sqrt{1 + \frac{20}{273}} = 341.9 \frac{\text{m}}{\text{s}}$$

$$t = \frac{d}{v} = \frac{4000 \text{ m}}{341.9 \frac{\text{m}}{\text{s}}} = 11.7 \text{ s} \rightarrow \text{Ans}$$

Reference: GEAS 02-10 – Sound

18. C. – 16 dB

Solution:

$$I = 10 \log \frac{I}{I_o} - 10 \log \frac{39I}{I_o}$$

$$I = -16 \text{ dB} \rightarrow \text{Ans}$$

Reference: GEAS 02-10 – Sound

19. B. 86 dB

Solution:

Displacement is the change in position.

$$I = \frac{P}{A} = \frac{2W}{4\pi(20\text{m})^2} = 3.98 \times 10^{-4} \frac{\text{W}}{\text{m}^2}$$

$$I = 10 \log \frac{I}{10^{-12} \frac{\text{W}}{\text{m}^2}}$$

$$I = 10 \log \frac{3.98 \times 10^{-4} \frac{\text{W}}{\text{m}^2}}{10^{-12} \frac{\text{W}}{\text{m}^2}}$$

$$I = 86 \text{ dB} \rightarrow \text{Ans}$$

Reference: GEAS 02-10 – Sound

20. A. 60 dB

Solution:

$$I = 10 \log \frac{I}{10^{-12} \frac{\text{W}}{\text{m}^2}}$$

$$I = 10 \log \frac{10^{-10} \frac{\text{W}}{\text{cm}^2} \left(\frac{100 \text{ cm}}{1 \text{ m}} \right)^2}{10^{-12} \frac{\text{W}}{\text{m}^2}}$$

$$I = 60 \text{ dB} \rightarrow \text{Ans}$$

Reference: GEAS 02-10 – Sound

21. A. 115 ft

Solution:

The time it travels one way is just 0.10 s.

$$d = vt = \left(1150 \frac{\text{ft}}{\text{s}} \right) (0.10 \text{ s})$$

$$d = 115 \text{ ft} \rightarrow \text{Ans}$$

Reference: GEAS 02-01 – Scalars and Vectors

22. B. 2.26×10^8

Solution:

$$n = \frac{c}{v}$$

$$1.33 = \frac{3 \times 10^8}{v}$$

$$v = 2.26 \times 10^8 \frac{\text{m}}{\text{s}} \rightarrow \text{Ans}$$

Reference: GEAS 02-12 – Nature of Light

23. B. 1.52



Solution:

$$n = \frac{c}{v} = \frac{3 \times 10^8 \frac{\text{m}}{\text{s}}}{123,000 \frac{\text{mi}}{\text{s}} \times \frac{1609 \text{ m}}{1 \text{ mi}}}$$

$$n = 1.52 \rightarrow \text{Ans}$$

Reference: GEAS 02-12 – Nature of Light

24. A. 55°

Solution:

For reflection, angle of incidence is always equal to the angle of reflection. Therefore, the angle of reflection is also 55 degrees.

Reference: GEAS 02-12 – Nature of Light

25. A. 0.55

Solution:

Use Thin-Lens Equation (it is assumed that the focal length is much greater than the thickness of the lens):

$$\frac{1}{f} = \frac{1}{i} + \frac{1}{o}$$

$$\frac{1}{-5 \text{ cm}} = \frac{1}{i} + \frac{1}{4 \text{ cm}}$$

$$i = -2.22 \text{ cm (virtual)}$$

f is + for a converging lens (thicker at the center than at the sides)

f is – for a diverging lens (thinner in the center than at the sides)

o is + if object is on the same side of the lens as the incident light (real object)

o is – if the object is on the other side

i is + if the image is on the side of the lens opposite the incident light (real)

i is – if the image is on the same side (virtual)

Use formula for Lateral Magnification:

$$m = -\frac{i}{o} = -\frac{(-2.2)}{4}$$

$$m = 0.55 \text{ (erect, smaller)}$$

Note:

+ value of m means image is erect

- value of m means image is inverted

m = 1 means object and image have same size

m < 1 means image is smaller

m > 1 means image is larger

Reference: GEAS 02-14 – Lenses

26. B. 2.5 diopters

Solution:

Nearsightedness (myopia) can be corrected by using a diverging lens while Farsightedness (hyperopia) can be corrected by using a converging lens.

$$D = \frac{1}{f}$$

$$\frac{1}{f} = \frac{1}{i} + \frac{1}{o}$$

$$D = \frac{1}{i} + \frac{1}{o} = \frac{1}{-1.25} + \frac{1}{0.3}$$

$$D = 2.5 \text{ diopters} \rightarrow \text{Ans}$$

Note: i is – because the image and object are on the same side of the lens (image must be virtual)

Reference: GEAS 02-14 – Lenses

27. B. 1.6×10^{-19}

Solution:

1 eV represents a very small quantity of energy:

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

Reference: GEAS 02-12 – Nature of Light

28. D. Wave frequency changes on entering a medium where the velocity of propagation is changed.

Solution:

When a wave transfers to another medium, the frequency will never change. Only the wavelength with change when the velocity changes.

Reference: GEAS 02-12 – Nature of Light

29. D. the index of refraction

Solution:

The index of refraction is the ratio of the speed of light in free space to the speed of light in the medium

Reference: GEAS 02-12 – Nature of Light

30. D. Blue light

Solution:

Energy is proportional to frequency. Between red, orange, green and blue, blue light has the highest frequency hence the most energy.

Note: $E = hf$

Reference: GEAS 02-12 – Nature of Light

31. C. Both angles are equal

Solution:

The Law of Reflection states that the angle of incidence and the angle of reflection should be equal.

Reference: GEAS 02-12 – Nature of Light

32. C. slower

Solution:

Light (and other forms of electromagnetic waves) travel fastest in vacuum. So in any other medium, it should be slower.

Reference: GEAS 02-12 – Nature of Light

33. B. refraction

Solution:



Refraction is the “bending” of light. This is the same reason why a swimming pool appears more shallow than it really is.

Reference: GEAS 02-12 – Nature of Light

34. B. visible light

Solution:

4000 to 7500 angstrom (or 400 to 750 nm) is the approximate range of wavelength values for visible light.

Reference: GEAS 02-12 – Nature of Light

35. B. angle of incidence is greater than the angle of refraction

Solution:

If light travels from low index to high index, it will bend towards the normal line. This means that the angle of incidence is greater compared to the angle of refraction.

Reference: GEAS 02-12 – Nature of Light

36. B. dispersion

Solution:

Dispersion is the process of the spreading or splitting of white light into the different colors of visible light.

Reference: GEAS 02-12 – Nature of Light

37. D. violet

Solution:

The high frequency (short wavelength) light is bent the most during refraction.

Reference: GEAS 02-12 – Nature of Light

38. A. Reflection

Solution:

Mirrors work because of the reflection phenomena.

Reference: GEAS 02-13 – Mirrors

39. B. Convex mirrors

Solution:

Convex mirrors are used and are popular because they give a wide view of the object.

Reference: GEAS 02-13 – Mirrors

40. C. Concave mirrors

Solution:

The only type of mirror that can magnify are converging mirrors (or concave mirrors)

Reference: GEAS 02-13 – Mirrors

41. B. Convex mirrors

Solution:

Convex mirrors are the ones used in the rear view mirrors of cars.

Reference: GEAS 02-13 – Mirrors

42. B. virtual image

Reference: GEAS 02-13 – Mirrors

43. A. Plane mirrors produce real images

Solution:

Plane mirrors ALWAYS produce upright and virtual images.

Reference: GEAS 02-13 – Mirrors

44. A. Convex

Solution:

For convex mirrors, the reflective side is the outer surface of the spherical mirror.

Reference: GEAS 02-13 – Mirrors

45. A. Convex

Solution:

Convex mirrors are used in parking areas to provide wide views.

Reference: GEAS 02-13 – Mirrors

46. A. one-half

Solution:

For spherical mirrors, the focal length is always half the radius.

Reference: GEAS 02-13 – Mirrors

47. A. the image is upright

Solution:

For concave mirrors, if the object is within the focal point or focal length, the image is upright, virtual and magnified.

Reference: GEAS 02-13 – Mirrors

48. C. magnification is greater than 1

Solution:

For concave mirrors, the magnification is always greater than 1 if the object is within the focal length.

Reference: GEAS 02-13 – Mirrors

49. D. magnification can be greater than or less than 1

Solution:

For concave mirrors, if the object is outside the focal length the image will be real and inverted. However, for magnification, it can be greater than 1 or less than 1.

Reference: GEAS 02-13 – Mirrors

50. B. negative

Solution:

Convex mirrors are diverging. Hence, the focal length is negative.

Reference: GEAS 02-13 – Mirrors